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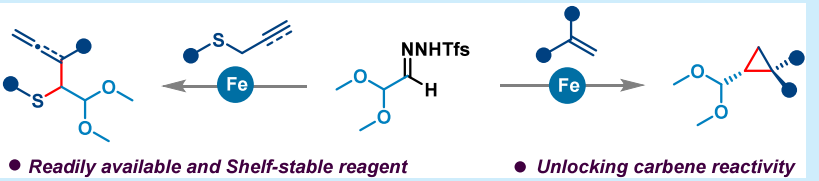
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Dimethoxyacetaldehyde-*N*-triftosylhydrazone: Preparation and Carbene Reactivity in Cyclopropanation and Doyle−Kirmse Reactions

Yifan Zhang,[#] Shuang Li,[#] Hongzhu Chen,[#] Wei Song,[#] Yongquan Ning,[*] Zeyun Liu, Yong Wu, Karunanidhi Murali, Paramasivam Sivaguru, and Xiaolong Zhang[*]

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ABSTRACT: Herein, we developed the new, powerful, and easy-to-handle chemical reagent, dimethoxyacetaldehyde-*N*-triftosylhydrazone (DMHz-Tfs), as a convenient in situ source of dimethoxydiazoethane under mild conditions. We demonstrate the carbene reactivity of DMHz-Tfs in iron-catalyzed cyclopropanation and Doyle−Kirmse reactions, providing access to diverse acetal functionalized cyclopropanes and homoallylic- and allenyl-sulfides at gram-scale with high stereoselectivity. DFT calculations elucidated the involvement of the most stable doublet spin state iron-carbene intermediate over other possible spin states.

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Since von Pechmann discovered diazomethane, the smallest diazo compound, in 1894,[1] advances in their synthesis and applications have expanded rapidly.[2] Although their synthesis requires stringent safety regulations,[3] the utility of diazo compounds as carbene precursors has become indispensable in modern organic synthesis.[4] The structural modifications of diazomethane to increase functionality and selectivity in chemical reactions have attracted more attention owing to its rich and tunable reactivity profile and its ability to prepare diverse functionalized chemical entities.[4c−i,5] α-Diazocarbonyl compounds[4c,d,5] and (*di*)aryl diazomethanes[6] were prominent in early applications. In recent decades, an array of α-functionalized diazoalkanes, including fluorinated moieties ($-CF_2H$,[7] $-C_nF_{2n+1}$[8−11]), $-CN$,[12] $-CHO$,[13] $-TMS$,[14] sulfone,[15] and phosphonate,[16] etc., have been practiced in diazo-based carbene chemistry. However, the synthesis and application of dimethoxydiazoethane remains largely untouched despite its potential to construct formylated building blocks (Figure 1A).

The history of dimethoxydiazoethane goes back to 1980 when Abdallah and Grée synthesized it from 2,2-dimethoxyethylamine in three steps and studied its reactivity as 1,3-dipoles to prepare acetal/formyl cyclopropanes and pyrazoles.[17] Later, Dehmlow and co-workers reported an improved synthesis protocol from the same starting materials without using N_2O_4 (Figure 1B).[18] Nonetheless, the need for multistep procedures, hazardous reagents, and acidic conditions limited its broad applicability. Moreover, the carbene reactivity of dimethoxydiazoethane has yet to be studied. Hence, the design and development of a new diazo surrogate that is able to release dimethoxydiazoethane species under mild conditions could greatly improve the reagent's synthetic accessibility, especially for large-scale applications. In pursuit of our continuous research efforts in the synthesis and applications of functionalized diazo alkanes derived from easily decomposable *N*-triftosylhydrazones,[19] we here introduce dimethoxyacetaldehyde-*N*-triftosylhydrazone (DMHz-Tfs) as the shelf-stable and safe-to-handle in situ precursor of dimethoxydiazoethane. The carbene reactivity of DMHz-Tfs was demonstrated through the iron-catalyzed cyclopropanation of alkenes and Doyle−Kirmse reaction with allyl and propargyl sulfides (Figure 1C), allowing access to diverse acetal functionalized cyclopropanes as well as homoallylic and allenyl sulfides in high yields with broad scope and excellent functional group compatibility.

Three dimethoxyacetaldehyde-*N*-sulfonylhydrazones with different sulfonyl groups could be readily prepared from commercially available 2,2-dimethoxyacetaldehyde in excellent yields (Scheme 1). These reagents are bench-stable crystalline solids that can be stored at ambient conditions for at least 3 weeks without significant degradation.

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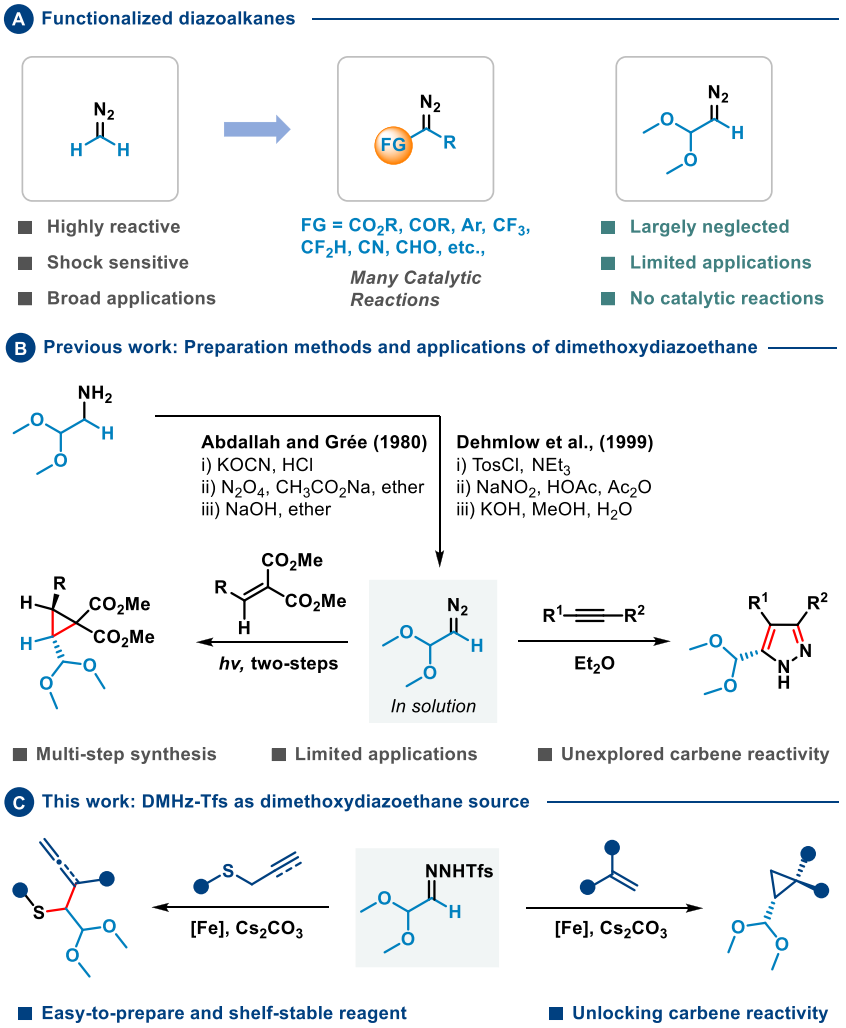
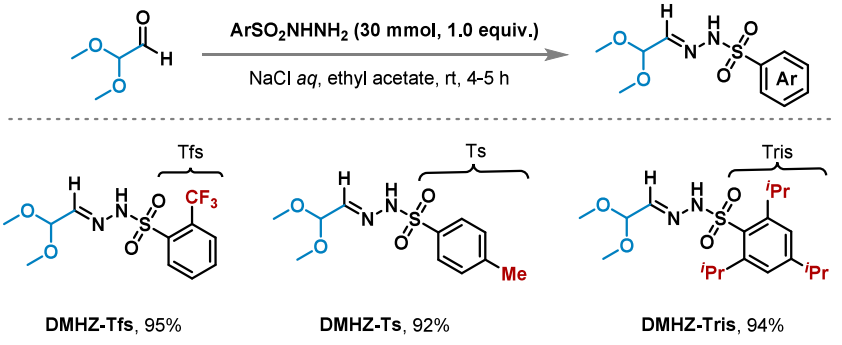


Figure 1. Dimethoxydiazoethane in situ derived from dimethoxyacetaldehyde-*N*-triftosylhydrazone (DMHz-Tfs) and its applications.

Scheme 1. Preparation of Dimethoxyacetaldehyde *N*-Sulfonylhydrazones[a]

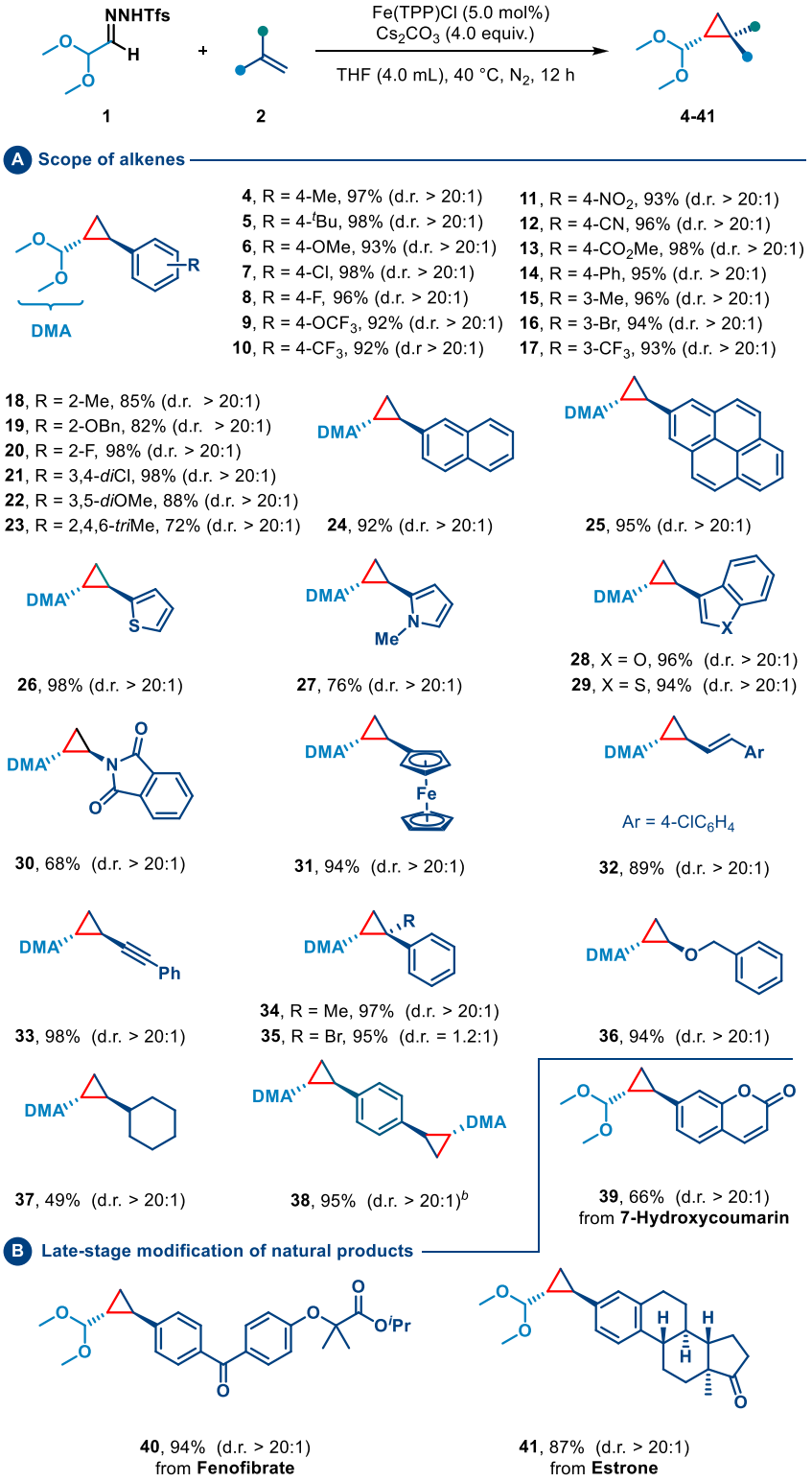


[a]Reaction conditions: 2,2-dimethoxyacetaldehyde (60 mmol, 2.0 equiv, 60% in water), $ArSO_2NHNH_2$ (30 mmol, 1.0 equiv), ethyl acetate (50.0 mL), and saturated NaCl aqueous (50.0 mL) at room temperature (rt). Isolated yields were shown.

Following extensive screening of the reaction conditions, it was observed that the Fe-catalyzed system was effective in the cyclopropanation of DMHz-Tfs (Supporting Information Table S1). With the optimized conditions in hand, we next examined the scope of alkenes for iron-catalyzed cyclopropanation with DMHz-Tfs (**1**, Scheme 2). Various styrenes bearing electron-donating (e.g., methyl, methoxy, *tert*-butyl, benzene, and benzyl ether) and electron-withdrawing (e.g., halogens, trifluoromethyl, nitro, cyano, and ester) groups were well tolerated, delivering the desired *trans*-cyclopropanes **4**−**20** in 72−98% yields. Notably, the electronic properties of the aryl rings had a negligible effect on the reaction efficiency. For *ortho*-substituted styrenes, the yield of the fluorine-substituted substrate was higher than methyl- and benzyl ether-substituted substrates (**20** vs **18** and **19**). Similarly, di- and tri-substituted styrenes as substrates could give the desired acetal cyclopropanes **21**−**23** in good yields. The reaction conditions were also applicable to fused aryl substrates (e.g., naphthalene and

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Scheme 2. Cyclopropanation Reaction of DMHz-Tfs[a]

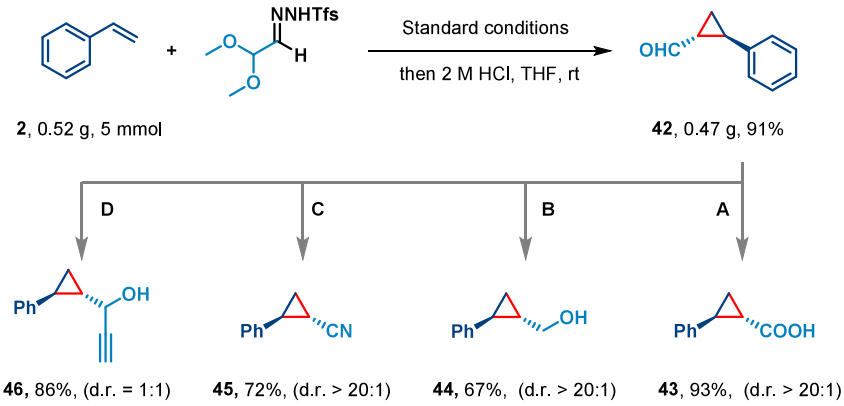


[a]Reaction conditions: DMHz-Tfs (2.0 equiv), alkene (1.0 equiv), Fe(TPP)Cl (5.0 mol %), Cs_2CO_3 (4.0 equiv), THF (4.0 mL) at 40 °C for 12 h under N_2. [b]DMHz-Tfs (4.0 equiv) and Cs_2CO_3 (8.0 equiv). Diastereomeric (d.r.) ratio was determined by ^1H NMR analysis of the crude reaction mixture. Isolated yields were shown.

pyrene), providing target products **24** and **25** in 92% and 95% yield, respectively. Alkenes embedded in electron-rich heterocycles, such as thiophene (**26**), *N*-methylpyrrole (**27**), benzofuran (**28**), benzothiophene (**29**), and phthalimide (**30**), converted to cyclopropanes without compromising reaction efficiencies. Notably, ferrocene was also compatible with the reaction to give an acetal cyclopropane (**31**). The reaction is highly regioselective, as only terminal cyclo-

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Scheme 3. Gram Scale Syntheses of 42 and Its Downstream Diversification[a]



[a](A) **42** (0.5 mmol) and $KHSO_5$ (0.5 mmol) in MeCN/H_2O at 35 °C for 24 h. (B) **42** (0.5 mmol) and $NaBH_4$ (0.6 mmol) in MeOH at 0 °C to rt for 2 h. (C) **42** (0.5 mmol) and $NH_2OH{\cdot}HCl$ (0.75 mmol) in *N*-methyl-2-pyrrolidone at 100 °C for 2 h. (D) **42** (0.5 mmol) and BrMgC≡CH (0.5 mmol) in THF at 25 °C for 12 h. Diastereomeric ratio (d.r.) was determined by ^1H NMR analysis of the crude reaction mixture.

propanation products **32** (89%) and **33** (98%) with conjugated diene and enyne as substrates were obtained. While α-methylstyrene (**34**) was cyclopropanated with high stereoselectivity, α-bromostyrene (**35**) led to only 1.2:1 stereoselectivity. In addition to aryl alkenes, aliphatic alkenes, such as enol ether and vinylcyclohexane, were amenable, furnishing the desired products **36** and **37** in 94% and 49% yields, respectively. The olefin substrate bearing other alkyl substituents afforded the corresponding cyclopropane products in very low yields (see Supporting Information). The reaction also tolerated *p*-divinylbenzene, which afforded the desired biscyclopropane **38** by doubling other reagents. Late-stage modification of complex molecules, such as 7-hydroxycoumarin, fenofibrate, and estrone derivatives, could also be achieved in good yields with excellent stereocontrol (**39**−**41**), demonstrating the potential for the discovery of more promising drug candidates.

To highlight the potential utility of this protocol, we scaled up the reaction of DMHz-Tfs and styrene to a 5 mmol scale, affording the acetal cyclopropane **3** in good yields without erosion of stereoselectivity. Then mild acid hydrolysis of **3** provided the formyl cyclopropane **42** in 91% yield, which could transform into structurally diverse cyclopropane scaffolds (Scheme 3). For example, oxidation of **42** with $KHSO_5$ led to the formation of carboxylic acid **43**, while it was reduced to primary alcohol **44** with $NaBH_4$. Cyclopropanecarbonitrile **45** was also obtained smoothly by treating **42** with hydroxylamine hydrochloride. Finally, the nucleophilic addition of ethynylmagnesium bromide to formyl cyclopropane **42** gave a diastereomeric mixture of propargyl alcohols **46** in 86% yield.

Next, we explored the reactivity of DMHz-Tfs in the Doyle−Kirmse reaction—an interesting [2,3]-sigmatropic rearrangement reaction to construct C−S and C−C bonds.[20] We chose DMHz-Tfs and allyl phenyl sulfide as model substrates to optimize the reaction conditions. After a series of experimentations, we were pleased to find that the target product **49** could be isolated in 95% yield when reacting **1** with allyl phenyl sulfide in the presence of Fe(TPP)Cl catalyst and Cs_2CO_3 base in THF at 40 °C for 12 h under nitrogen atmosphere (see Supporting Information Table S2 for details).

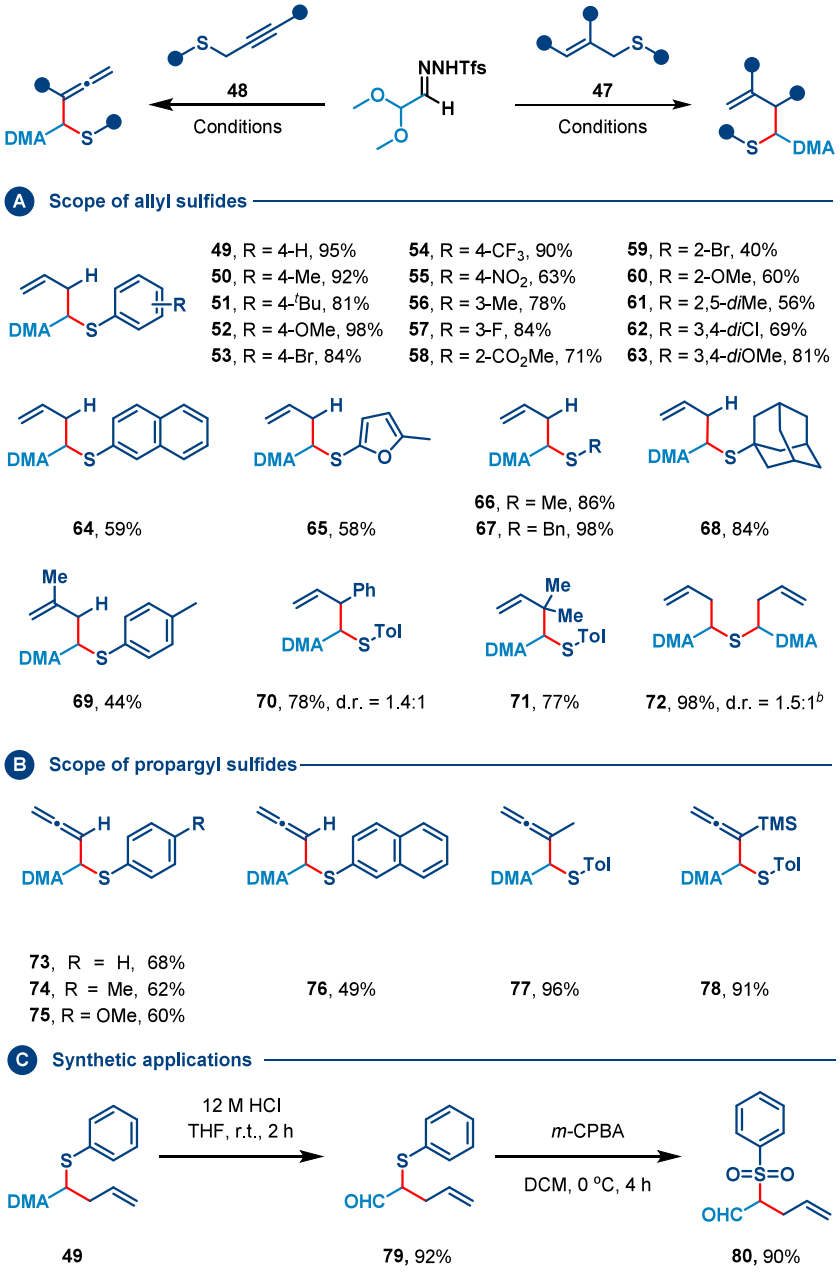
Under the optimized conditions, the generality of the reaction was then examined (Scheme 4A). A wide range of aryl allyl sulfides containing either electron-donating or electron-withdrawing substituents at the *para- or meta*-positions of the aromatic ring were compatible, affording corresponding products **50**−**57** in 63 to 98% yields. The steric hindrance on the aromatic ring impacted the product yields, as *ortho*-substituted aryl allyl sulfides produced slightly lower yields (**58**−**60**). The reaction was also amenable to disubstituted aryl allyl sulfides to deliver the desired products in moderate to good yields (**61**−**63**). Allyl sulfides bearing electron-rich naphthalene and furan rings smoothly participated in the process, delivering **64** and **65** in 59% and 58% yields, respectively. Besides aryl allyl sulfides, methyl, benzyl, and adamantyl allyl sulfides were also suitable for this conversion, which could react with DMHz-Tfs to obtain corresponding products **66**−**68** in moderate to good yields. Similarly, the reaction using 2-methylallyl, 1-phenylallyl, and 1,1-dimethylallyl aryl sulfides was also attempted, furnishing the desired products in moderate to good yield (**69**−**71**). It is worth noting that diallyl sulfide can effectively engage in the reaction, yielding the desired bilateral [2,3]-sigmatropic rearrangement product **72** (98% yield and d.r. = 1.5:1) by doubling other reagents.

We then expanded the scope of this reaction to propargyl sulfides **48** (Scheme 4B), which offered a convenient route to functionalized allenes.[21] Aryl propargyl sulfides bearing a methyl or methoxy group at the *para*-position of the aryl ring and naphthyl propargyl sulfide could be converted smoothly into the corresponding allenyl sulfides **73**−**76** in moderate to good yields (49−68%). The reaction could proceed effectively with 1-methyl or 1-trimethylsilyl substituted alkynyl sulfides, which delivered target products **77** and **78** in 96% and 91% yield, respectively. Finally, we explored the versatility of the acetal-substituted homoallyl sulfide product **49**. The mild acid hydrolysis of **49** produced the corresponding aldehyde analog **79** (92% yield), which then was oxidized in the presence of *m*-CPBA to give sulfone derivative **80** in 90% yield (Scheme 4C).

To gain insight into the reaction mechanism of cyclopropanation, we performed density functional theory (DFT) calculations on a model reaction of DMHz-Tfs with styrene (Figure 2). The Fe catalyst could exhibit three distinct spin states.[22] First, **1** decomposed readily in the presence of Cs_2CO_3 generating the diazo species **1a**.[23] Subsequent interaction of **1a** with Fe(TPP)Cl generates stable doublet electrophilic iron-carbene intermediate **Int2** with the elimi-

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Scheme 4. Doyle−Kirmse Reaction of DMHz-Tfs[a]



[a]Reaction conditions: DMHz-Tfs (3.0 equiv), sulfide (1.0 equiv), Fe(TPP)Cl (5.0 mol %), Cs_2CO_3 (6.0 equiv), THF (4.0 mL) at 40 °C for 12 h under N_2. [b]DMHz-Tfs (6.0 equiv) and Cs_2CO_3 (12.0 equiv). Diastereomeric ratio (d.r.) was determined by ^1H NMR analysis of the crude reaction mixture. Isolated yields were shown.

nation of dinitrogen *via* doublet transition state ^{2}**TS1** ($\Delta G^{\ddagger}$ = 1.1 kcal/mol). Compared to those of ^{2}**TS1**, the energy barriers for forming quartet transition state ^{4}**TS1** ($\Delta G^{\ddagger}$ = 4.0 kcal/mol) or sextet transition state ^{6}**TS1** ($\Delta G^{\ddagger}$ = 9.5 kcal/mol) are significantly higher. Subsequent electrophilic concerted addition of iron carbene **Int2** to styrene **2** generates the cyclopropane *via* transition state ^{2}**TS2** with an energy barrier of 5.7 kcal/mol, which is the rate-determining step. Similarly, the energy barriers of the addition process for the quartet and sextet transition states are much higher (^{4}**TS2**: $\Delta G^{\ddagger}$ = 12.1 kcal/mol; ^{6}**TS2**: $\Delta G^{\ddagger}$ = 17.4 kcal/mol) than the doublet transition state and are thus excluded.

To probe the origin of the diastereoselectivity, we performed an interaction region indicator (IRI) analysis of the diastereoselectivity-determining transition state ^{2}**TS2** (Figure 2B). Our calculations indicate that the *anti*-selective transition state has a stronger $\pi\cdots\pi$ interaction between the C=C bond and the TPP ligand, resulting in a lower energy barrier ($\Delta G^{\ddagger}$ = 5.7 kcal/mol) than the *syn*-selective transition state ($\Delta G^{\ddagger}$ = 10.8 kcal/mol). Thus, the *anti*-cyclopropane formation is far preferable to that of *syn*-cyclopropane (see Supporting Information for details). On the other hand, according to our calculations for Doyle−Kirmse reaction, the nucleophilic sulfur attack on iron carbene **Int2** forms the zwitterionic intermediates **Int3** *via* **TS2**, which then undergoes a rate-determining [2,3]-sigmatropic rearrangement to furnish the final product while regenerating the Fe(TPP)Cl catalyst. We also considered free ylide,[24] which was excluded because the

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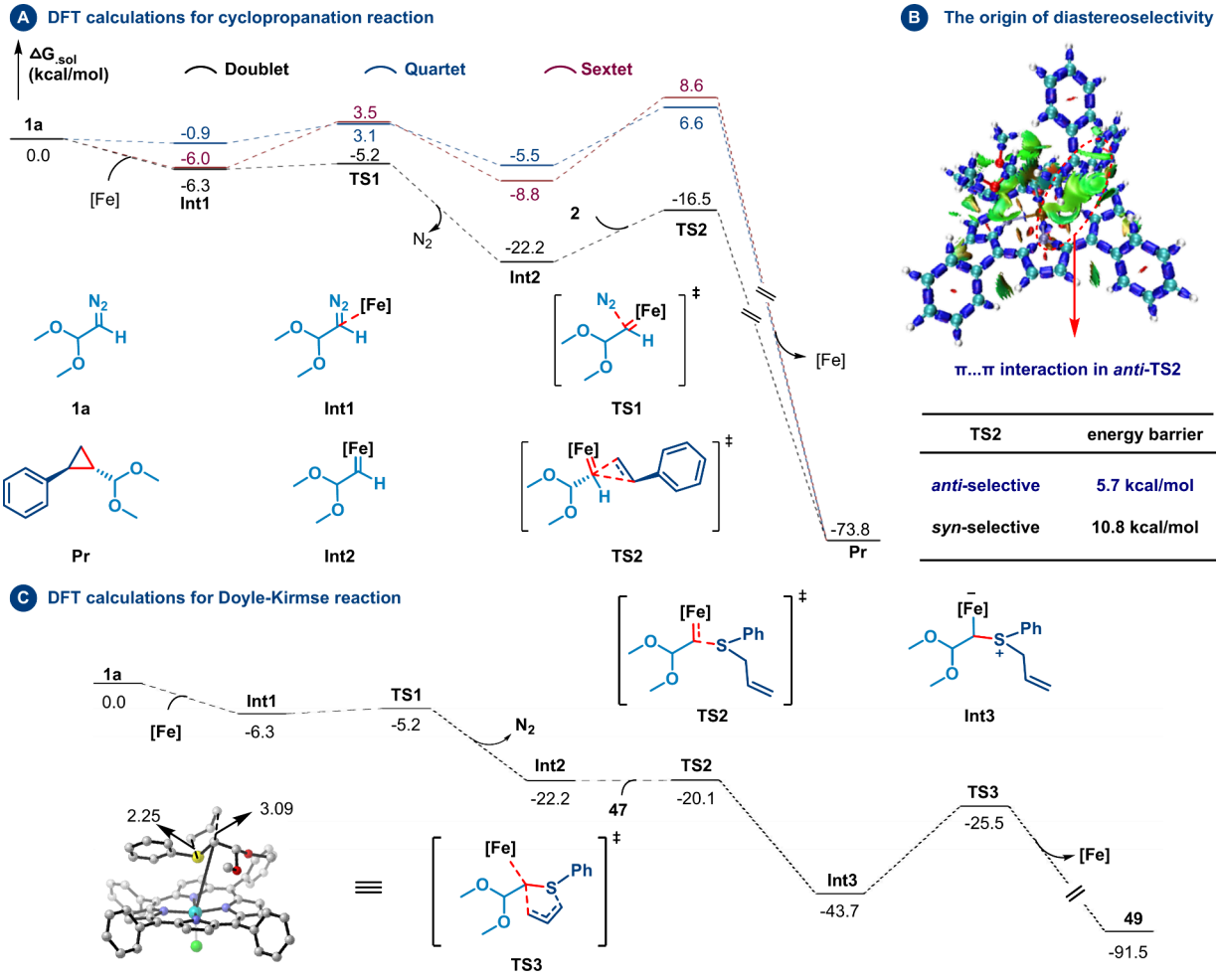


Figure 2. DFT calculations at the SMD(THF)-B3LYP/6-31G(d,p)-SDD(Fe) level of theory; energies are given in kcal/mol.

energy barrier of free ylide pathway (57 kcal/mol) is much higher than that of metal-bound ylide (18.2 kcal/mol, see Supporting Information for details).

In this work, we developed a new chemical reagent, DMHz-Tfs, which could slowly release dimethoxydiazoethane in situ under mild conditions. Its carbene reactivity was established in iron-catalyzed cyclopropanation and Doyle−Kirmse reactions, enabling access to various acetal functionalized cyclopropanes and homoallyl- and allenyl-sulfides. The broad applicability of the developed protocol was highlighted by the late-stage modification of complex bioactive molecules, gram-scale synthesis, and structural diversification of acetal functionalized products. DFT studies disclosed that the doublet spin state of the iron catalyst enabled the carbene transfer process. Given its ease of preparation, high stability, and operational safety, DMHz-Tfs are expected to have a high potential for carbene transfer reactions.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00223.

Experimental details, computational details, characterization data, NMR spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Yongquan Ning − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*; orcid.org/0000-0001-8791-6254; Email: ningyq508@nenu.edu.cn

Xiaolong Zhang − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*; Email: zhangxl226@nenu.edu.cn

Authors

Yifan Zhang − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*

Shuang Li − *Forestry College of Beihua University, Jilin 132013, PR China*

Hongzhu Chen − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*

Wei Song − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*

Zeyun Liu − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*

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Yong Wu − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China;* orcid.org/0000-0002-6806-1817
Karunanidhi Murali − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China*
Paramasivam Sivaguru − *Department of Chemistry, Northeast Normal University, Changchun 130024, PR China;* orcid.org/0000-0003-2225-0054

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c00223

Author Contributions

#YZ, SL, HC, and WS contributed equally to this work.

Notes

The authors declare no competing financial interest.

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